

NEET CBT 2027 Simulator — Marketing Site

Academic Specification — Automated Architectural Analysis via code2paper

Abstract — The repository hosts the static marketing site for the NEET CBT 2027 Simulator, a free, offline-first progressive web application that replicates the National Testing Agency (NTA) computer-based test interface for the NEET UG medical entrance examination. The site is a single-file HTML document (423 lines, 23.5 KB) with embedded CSS and JSON-LD structured data, deployed via GitHub Pages with no build pipeline. The repository contains 9 files across HTML, Markdown, XML, plain-text, and raster image assets; source lines total approximately 455.

1. System Overview

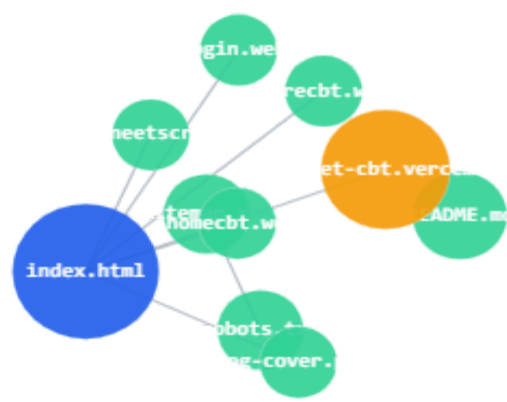
Metric	Value
Total files	9
Text files	4 (index.html, README.md, robots.txt, sitemap.xml)
Image assets	5 (.webp screenshots, og-cover.png)
Source lines (text)	~455
Build step	None (static)
Hosting	GitHub Pages (autodeploy on push)
Dependencies	Zero

Language breakdown: HTML (index.html, 423 lines), Markdown (README, 17 lines), XML (sitemap.xml, 9 lines), plain text (robots.txt).

2. Architecture & Component Structure

The site is a linear one-page document. The dependency graph below is interactive — drag nodes, hover to highlight connections. It maps every file to what it references, including the external simulator app.

entrypoint (index.html)
asset / config
external app (neet-cbt.vercel.app)
edge = reference / link



drag to move • scroll to zoom • hover to trace

Structural layout

```
index.html
├─ <head>
│   └─ primary/OG/Twitter meta + <style> design system
├─ <body>
│   └─ .topbar      - announcement strip ("CBT begins 2027")
│   └─ header.main  - brand + nav (Why/Features/How/FAQ anchors)
│   └─ .hero        - headline, lead, dual CTA, stats badges, CSS mock-UI
│   └─ #why          - 4-card rationale grid
│   └─ #features     - 6-feature grid (NTA interface, timer, PWA, PYQ, privacy)
│   └─ #how          - 3-step + 4-screenshot gallery
│   └─ #faq          - 6-item <details> disclosure
│   └─ footer        - links + NTA trademark disclaimer
└─ 3 × JSON-LD blocks - WebApplication, FAQPage, BreadcrumbList
```

Dependency edges

- index.html → testneetscreen.webp , homecbt.webp , login.webp , scorecbt.webp (gallery, lazy-loaded)
- index.html → og-cover.png (social meta)
- index.html → neet-cbt.vercel.app (external simulator app, 2 CTAs + footer)
- robots.txt → sitemap.xml

3. Core Module Specifications

index.html — single-file landing page

- **Role:** the entire marketing site.
- **Design system:** CSS custom properties in `:root` (navy, accent gold, blue, paper), responsive at 820px.
- **CTA strategy:** two "Launch the Simulator" buttons funnel to the app; "Why the change?" scroll anchor.
- **Trust architecture:** stat badges, six FAQ disclosures, non-affiliation disclaimer.
- **SEO:** descriptive meta, OG + Twitter cards, canonical, three JSON-LD entities, theme-color.

README.md

Confirms GitHub Pages hosting and image conventions (WebP at 2048×3640).

robots.txt / sitemap.xml

Allow all crawlers; sitemap with monthly change frequency, dated 2026-08-10.

4. Mathematical Formalization & Data Flow

Let F be the set of files and $\ell(f)$ the line count of file f .

Asset size footprint (bytes):

Data flow is unidirectional and static: the browser fetches `index.html` plus lazily-loaded gallery images; JSON-LD blocks are parsed by crawlers, not executed. No runtime computation occurs here.

5. Implementation Trade-offs & Limitations

1. **Single-file simplicity vs. maintainability:** zero build, zero deps, at the cost of a 423-line file.
2. **Isolation from the product:** marketing site and simulator are separate repos.
3. **Static screenshots:** WebP captures cannot reflect future UI changes automatically.
4. **Scope:** no test suite or CI beyond Pages autodeploy; appropriate for a static page.